2025-spring Introduction to Quantum Information and Quantum Computing

HW # 3 (Due 5/21 midnight = 5/22 00:00 am). Hand in as a single pdf file through ETL. For Qiskit problems: Pasting only the result (output) of the code (perhaps as a picture) will suffice—no need to hand in the coding itself unless explicitly specified. Also, running on an actual backend is recommended if possible, but the simulation result is also Okay.

1. In the past lectures on entanglement, we skipped the **PPT criterion.** Now we want to revisit it. The idea simply relies on the fact that the transpose operation is positive but not completely positive. Remember when we showed this in the class, we applied $I_A \otimes T_B$ to an entangled state where A and B denote sussystems.

A. Show that the transpose operation produces valid density matrix (that is positive matrix) when the above form $I_A \otimes T_B$ is applied to a separable state.

B. Motivated from A, the PPT criterion says that if application of $I_A \otimes T_B$ to the given state produces an operator having at least one negative eigenvalue, then the given state is entangled. We already checked this for the maximally entangled Bell states. Try with the **Werner state** $\rho_W = p\left|\beta_{00}\right\rangle\left\langle\beta_{00}\right| + (1-p)\frac{\mathbb{I}}{2} \otimes \frac{\mathbb{I}}{2}$ and mention the condition for which is state is entangled.

C. Consider a situation where a dephasing channel $\mathcal{E}_{deph}(\rho) = (1-p)\rho + p\hat{\sigma}_Z \rho \hat{\sigma}_Z$ is applied to one of the two-qubit entangled Bell state. Apply PPT condition on this state (dephased Bell state) and mention the condition for entanglement.

2. First off, read and learng yourself the section 9.5.1 of the textbook where quantum teleportation for mixed states is discussed. Then consider the following problem. In the quantum teleportation protocaol, Alice wished to transmit a pure state $\rho = \left|\psi\right\rangle\left\langle\psi\right|$. Suppose that the entangled pair shared by Alice and Bob is affected by some noise $I \otimes E_{deph}\left|\beta_{00}\right\rangle\left\langle\beta_{00}\right|$ where the dephasing channel E_{deph} is defined above. Compute the average fidelity $\bar{F}$ of the teleportation protocol.

3. Read and learn yourself. Example 9.11 of the textbook. This example provides a be며샤례 interpretation of the Choi-Jamiolkowski isomorphism.

4. **Qiskit –** Do DIY problem of the exercise 6.8 of the lecture note (making automatic loops for measuring the excited state probability at various waiting times, and making a plot of probability vs. waiting time. Then obtain T_1 from the fit).

5. Do exemple 6.14 of the lecture note.

6. **Textbook exercise. 7.1** You can of course refer to the solution provided in the textbook. However, I trust that you do not just copy and paste it.

7. **Textbook exercise. 7.2** Same as above

8. **Textbook exercise. 7.5** Same as above

9. **Quantum process tomography.** We skipped QPT in the class, but I believe you can do this. This is from another (perhaps more famous) textbook. From Nielsen & Chuang p393, single qubit process tomography can be done easily through the following steps. With the following information and input states $|0\rangle, |1\rangle, |+\rangle, |-\rangle$, calculate process matrix (chi-matrix below) of the Hadamard gate.

Box 8.5: Process tomography for a single qubit

The general method of process tomography can be simplified in the case of a one qubit operation to provide explicit formulas which may be useful in experimental contexts. This simplification is made possible by choosing the fixed operators $\tilde{E}_i$ to have commutation properties which conveniently allow the χ matrix to be determined by straightforward matrix multiplication. In the one qubit case, we use:

$$E_0 = I \tag{8.169}$$
$$\tilde{E}_1 = X \tag{8.170}$$
$$\tilde{E}_2 = -iY \tag{8.171}$$
$$\tilde{E}_3 = Z. \tag{8.172}$$

There are 12 parameters, specified by χ, which determine an arbitrary single qubit quantum operation $\mathcal{E}$.
These parameters may be measured using four sets of experiments. As a specific example, suppose the input states $|0\rangle$, $|1\rangle$, $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ and $|-\rangle = (|0\rangle + i|1\rangle)/\sqrt{2}$ are prepared, and the four matrices

$$\rho_1' = \mathcal{E}(|0\rangle\langle 0|) \tag{8.173}$$
$$\rho_4' = \mathcal{E}(|1\rangle\langle 1|) \tag{8.174}$$
$$\rho_2' = \mathcal{E}(|+\rangle\langle +|) - i\mathcal{E}(|-\rangle\langle -|) - (1-i)(\rho_1' + \rho_4')/2 \tag{8.175}$$
$$\rho_3' = \mathcal{E}(|+\rangle\langle +|) + i\mathcal{E}(|-\rangle\langle -|) - (1+i)(\rho_1' + \rho_4')/2 \tag{8.176}$$

are determined using state tomography. These correspond to $\rho_j' = \mathcal{E}(\rho_j)$, where

$$\rho_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \tag{8.177}$$

$\rho_2 = \rho_1 X$, $\rho_3 = X\rho_1$, and $\rho_4 = X\rho_1 X$. From (8.156) and Equations (8.169)–(8.172) we may determine β, and similarly ρ_j' determines λ. However, due to the particular choice of basis, and the Pauli matrix representation of $\tilde{E}_i$, we may express the β matrix as the Kronecker product $\beta = \Lambda \otimes \Lambda$, where

$$\Lambda = \frac{1}{2}\begin{bmatrix} I & X \\ X & -I \end{bmatrix}, \tag{8.178}$$

so that χ may be expressed conveniently as

$$\chi = \Lambda \begin{bmatrix} \rho_1' & \rho_2' \\ \rho_3' & \rho_4' \end{bmatrix} \Lambda, \tag{8.179}$$

in terms of block matrices.